Area Moments of Inertia
Transformation to Rotated Axes

Given or find for an Area (about its x and y axes): Ix, Iy, and Ixy

(Note: If Ixy = 0, then this Area's Ix and Iy are principal axes (I_{MAX} and I_{MIN})).

To find Iu, Iv and Iuv about an arbitrary rotated coordinate system (at angle θ):

1. Equations in terms of θ. (Can use the Iu equation for both I_θ and $I_{\theta+90}$, just put the correct angle in the equation; so technically do not need the Iv eqn)

1(a) $$\boxed{I_u = I_x cos^2\theta + I_y sin^2\theta - 2I_{xy}\sin\theta\cos\theta}$$

1(b) $$\boxed{I_v = I_x sin^2\theta + I_y cos^2\theta + 2I_{xy}\sin\theta\cos\theta}$$

1(c) $$\boxed{I_{uv} = I_{xy}(cos^2\theta - sin^2\theta) + \left(I_x - I_y\right) sin\theta cos\theta}$$

For Iuv easier to use the 2θ form (2(c)) of the equation below.

2. Equations in terms of 2θ. (Slightly easier to calculate, no need to square sin and cos terms)

2(a) $$\boxed{I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} cos2\theta - I_{xy} sin2\theta}$$

2(b) $$\boxed{I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} cos2\theta + I_{xy} sin2\theta}$$

2(c) $$\boxed{I_{uv} = \frac{I_x - I_y}{2} sin2\theta + I_{xy} cos2\theta}$$

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3. We are especially interested in the Principal Moments of Inertia (I_{MAX} and I_{MIN}) and the angle θ_p from the x-y coordinate system.

$$\boxed{I_{MAX\ or\ MIN} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}}$$

I_{MAX} and I_{MIN} are located at axes oriented at θ_{p1} and $\theta_{p2} = \theta_{p1} + 90$ measured from the x orientation.

θ_p is the angle from the x orientation to the nearest principal I, either I_{MAX} or I_{MIN}.

$$\boxed{tan2\theta_p = \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}}$$

Note that Ixy is a measure of the skew of the principal axes from the original x-y orientation. If $I_{xy} = 0$, then $\theta_p = 0$.